

A Global 3° Periodic Structure in the UNESCO World Heritage

Longitude Distribution: Domed Monuments, the Babylonian Beru, and a Levantine Harmonic Grid

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Abstract

This exploratory study documents a global 3° periodic structure in the UNESCO World Heritage Cultural and Mixed longitude distribution ($N = 1011$). No timestamped pre-registration was filed, and all results are treated as exploratory.

Every longitude of the form $x.18^\circ\text{E}$ (120 equally spaced anchors) yields the maximum observed count of sites within 6.7 km of a harmonic node (Tier-A+, against a 4% geometric null). The 3° spacing equals 0.1 beru, the smallest integer sub-division of the Babylonian angular unit (MUL.APIN, c. 1000 BCE). Gerizim's longitude falls within this phase band, initiating the analysis (§1.1).

Domed and spherical monuments ($N = 90$) show 12.2% Tier-A+ (permutation $Z = 3.20$, $p = 0.004$; Bonferroni $p_{\text{adj}} = 0.004$, $k = 2$). A+-bearing harmonics host $4.19\times$ more total sites than non-A+-bearing harmonics (permutation $p = 0.0033$). Leave-one-out checks confirm no single site drives either result.

Geographic concentration of Eurasian civilizations in longitude bands aligned with beru harmonics cannot be excluded on this corpus alone. The dome enrichment survives all three simulation nulls and exceeds the expectation from monuments at the same longitude footprint (Null C, §3.1), but the within-dome bootstrap confirms that domed monuments' own longitudes encode the enrichment (Null A, ns). The primary contribution is documentation of a reproducible quantitative regularity. The metrological interpretation awaits independent historical confirmation.

Keywords: archaeometry, ancient units of measurement, stupa, spatial statistics, heritage site geography, quantitative archaeology, harmonic grid, cuneiform metrology

1 Introduction

The UNESCO World Heritage Cultural and Mixed corpus ($N = 1011$) appears to contain a global 3° periodic structure in its longitude distribution. Every longitude of the form $x.18^\circ\text{E}$ produces the maximum observed Tier-A+ count when used as an angular reference (sites within $\lesssim 6.7$ km of a harmonic node, against a 4% geometric null). The 3° spacing equals the smallest integer sub-division of the Babylonian *beru* unit ($0.1 \text{ beru} = 3^\circ = 3 \text{ uš}$, MUL.APIN, c. 1000 BCE). The analytic consequences of the Gerizim phase-coincidence that initiated this study are defined in §1.1.

1.1 Circularity Containment Protocol

This subsection is the single canonical statement of the circularity problem and its analytic consequences. All later mentions are back-references to this section.

How the circularity arose. This study began with the hypothesis that the Babylonian *beru* unit might be detectable as a longitude-spacing signal in the UNESCO corpus, using Mount Gerizim (35.269°E) as the reference anchor. Running the global anchor sweep revealed that Gerizim's longitude falls within the $x.18^\circ\text{E}$ phase band. If the sweep had revealed a different optimal phase, this paper would not exist in its current form.

What the circularity contaminates.

- The Levantine corridor statistics (Tests 1, 2, 3 as computed from Gerizim), since reporting them was conditioned on observing the phase alignment.
- The $x.18^\circ\text{E}$ framing, which was identified in the sweep, not predicted independently.
- The Rayleigh Test B, whose near-perfect R follows mathematically from the A+ classification criterion (§3.3).
- All narrative emphasis on the Levantine corridor as historically motivated.

What the circularity does *not* contaminate.

- *The existence of a 3° periodic structure in the UNESCO longitude file.* Any investigator running an unanchored 120-phase scan will recover it without reference to Gerizim.
- *The dome enrichment (Test 2).* All 120 $x.18^\circ\text{E}$ anchors produce identical A+ counts for the dome population.
- *The cluster asymmetry (Test 3).* This is a full-corpus result requiring no keyword filter.
- *The unit sensitivity sweep.* The sharp collapse at $\pm 0.3\%$ of the canonical spacing is a property of the corpus structure, not of the choice of anchor.

The protocol. *This identification problem cannot be closed by any statistical method applied to this corpus.* There is no way to demonstrate after the fact whether the $x.18^\circ\text{E}$ phase was discovered before or after Gerizim was chosen as the anchor; even a time-stamped archive cannot resolve this because the internal sequence of analysis steps is unverifiable by any external party. Accordingly:

- (a) All results are treated as *exploratory* without exception.
- (b) The Bonferroni correction is applied as a *descriptive bound* on effect magnitude.
- (c) The Levantine corridor statistics are reported as the motivated entry point, not as independent evidence.
- (d) Downstream sections back-reference this section rather than re-stating the argument.

Why this design avoids the main pitfalls of archaeoastronomy. Prior metrological investigations were rightly criticized for investigator-assembled site lists (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). Three features distinguish this study. First, the UNESCO site list was compiled without reference to this hypothesis and is fixed. Second, the angular interval (0.1 beru) was selected on metrological grounds, not tuned post-hoc. Third, tier boundaries and keyword sets were specified before any beru distances were computed.

Structure of the paper. Section 1.2 provides background on the beru unit (subsection of Introduction). Section 2 describes data, classification, and multiple-comparisons protocol. Section 3 presents results. Section 4 reports robustness checks. Section 5 presents discussion, including limitations (§4.7) and the conclusion (§5.5). The Appendix contains the keyword audit.

1.2 Background

1.2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), dated on broad grounds to the late second or early first millennium BCE. In this system the unit divided the ecliptic into twelve equal segments and served for both stellar and terrestrial angular measurement. The terrestrial use is attested explicitly in MUL.APIN II ii 43–iii 15, which converts beru of daylight into ground distances (Hunger & Steele, 2019, pp. 2–15). The sub-division into 30 *uš* (1 *uš* = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to exactly 3 *uš*.

The beru also functioned as a unit of territorial description. Royal inscriptions record inter-regional distances in beru between major cities (Neugebauer, 1975; Robson, 2008). On the Babylonian *Imago Mundi*, distances between the “islands” beyond the encircling Salt Sea are marked as multiples of beru. The *kudurru* (boundary stone) tradition records territorial extents in cuneiform, and many *kudurru* were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987). This combination of a fractional angular unit at cosmological scales and temple-deposited territorial records provides the documentary basis for testing whether a 0.1-beru harmonic spacing is detectable in the longitude distribution of major monuments. This study tests the statistical pattern only. The mechanism by which any such correspondence could arise is not documented and is outside the scope of this paper.

1.2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested history. Thom’s “megalithic yard” (Thom,

1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Powell’s treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference. More recently, Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties.

This study departs from prior work in three respects. First, the site list is externally curated (the UNESCO World Heritage List), eliminating selection bias. Second, the test is one-dimensional (longitude only), yielding a well-defined geometric null (4%) with no free parameters. Third, the analysis is fully computational and publicly archived (Tenenbaum, 2026).

1.2.3 The Anchor: Mount Gerizim

The circularity arising from Gerizim’s longitude coinciding with the optimal phase band is defined in §1.1.

Mount Gerizim (35.269°E) is the primary reference anchor because of its documented *tabbur ha-aretz* (“navel of the earth”) role in Samaritan cosmographic tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), analogous to Babylon’s meridian in cuneiform astronomical records (Hunger & Steele, 2019). No optimization across alternative Levantine anchors was performed.

Gerizim’s tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO inscribed corpus.¹ Jerusalem (35.2317°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member. It is self-excluded when used as anchor and produces the same A+ count (§4.3). The enrichment belongs to the 4.1 km corridor, not to the specific choice of anchor within it. Mecca reaches only $p \approx 0.057$ (ns).

1.2.4 Interpretive Framework

This paper is a pattern-detection study, not a historical attribution. Three classes of explanation are evaluated throughout.

1. **Geographic concentration.** Major civilizations that produced UNESCO-ranked monuments developed in Eurasian longitude bands that align with 0.1-beru harmonics. The dedicated geographic-concentration simulation (§3.1) shows this remains a viable first-level account.
2. **Metrological transmission.** A 3° (0.1-beru) angular reference system anchored near the Levant informed the placement of major foundations. No mechanism is documented.
3. **Unmodeled statistical artifact.** A systematic bias in UNESCO curation, geocoding, or the discrete longitude distribution produces the apparent periodicity. An artifact account must additionally explain the dome enrichment and the cluster asymmetry.

¹UNESCO World Heritage Centre, *Mount Gerizim and the Samaritans*, Tentative List ref. 5706 (State of Palestine, submitted 02 April 2012; criteria (iii)(vi); coordinates N 32°12′44″ E 35°16′08″). <https://whc.unesco.org/en/tentativelists/5706/>. Archived rendered copy: https://ia601602.us.archive.org/33/items/unesco_5706_rendered/unesco_5706_rendered.html.

Adjudicating between explanations 1 and 2 requires independent corpora.

2 Data and Methods

2.1 Corpus Construction

The UNESCO World Heritage XML dataset contains 1248 inscribed nominations across all categories. Two outcome-blind filters yield the analysis corpus: (1) Natural-only sites are excluded (-235 , retaining Cultural and Mixed); (2) entries lacking geocoordinates are excluded (-2). The result is $N = 1011$ sites with valid longitudes.

2.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E , from Palestine’s UNESCO tentative list (ref. 5706, 2012).² Gerizim is not on the inscribed World Heritage List and does not appear in the test-site corpus. Jerusalem is also excluded in both sweeps (§1.2.3).

For a site at longitude λ , the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \text{round}(B, 0.1)|$. The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 6.7$ km for Tier-A+) are equatorial approximations computed as $\delta \times 30 \times 111$ km/ $^\circ$, becoming smaller at higher latitudes. This makes the angular criterion *conservative* at high latitudes. A geodesic sensitivity check confirms that applying a strict physical threshold would increase the A+ count to 70, strengthening all primary results. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§4.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 40.4 ± 6.2 expected A+ ($\sim 4.0\%$), confirming the geometric null is not materially inflated by longitude clumping.

2.3 Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested is 0.1 beru (3°), the smallest integer sub-division of the beru attested in cuneiform metrology, equalling exactly 3 *uš* (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). It was selected because it is the finest metrologically attested integer sub-division that yields a tractable test (120 harmonics, 4% geometric null). Coarser spacings (0.5 beru, 1 beru) are uninformative; finer sub-divisions (0.02 beru) do not produce the signal (Table 10). No spacing was selected post-hoc to maximize the result.

The 0.05-beru spacing (6°) also reaches significance (Table 10). This is structurally expected because every 0.10-beru harmonic is also a 0.05-beru harmonic, making the 0.05-beru result a coarser sub-sampling of the same periodicity, not a separate signal.

2.4 Tier Classification (fixed before data analysis)

Sites are classified by δ , the angular deviation from the nearest 0.1-beru harmonic. All tier boundaries and test populations were defined before any beru values were computed.

- **Tier A++:** $\delta \leq 0.0002$ beru ($\lesssim 0.67$ km). $P_0 = 0.4\%$. Descriptive only, not tested.

²See footnote in §1.2.3 for full citation and archived URL.

- **Tier A+:** $\delta \leq 0.002$ beru ($\lesssim 6.7$ km). $P_0 = 4\%$. Primary test threshold.³
- **Tier A:** $\delta \leq 0.010$ beru ($\lesssim 33$ km). $P_0 = 20\%$.
- **Tier B:** $\delta \leq 0.050$ beru.

2.5 Domed and Spherical Monuments (Test 2)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text. All keywords are treated identically (positive whole-word match, no context gating), making the population mechanically determined by the keyword list. Final population: $N = 90$.

A context-validated robustness check (Test 2x) applies sentence-level architectural confirmation, reducing the population to $N = 83$ (7 sites rejected, e.g. Uluru “rock domes,” Hiroshima “Genbaku Dome,” full list in Appendix A.1). The validated version yields $p = 0.0005$, $3.31\times$ enrichment, stronger than the raw sweep ($p = 0.0009$, $3.06\times$), consistent with rejected sites being non-architectural false positives that dilute the signal. Because the validation rules were developed after the initial sweep, Test 2x is a post-hoc robustness check excluded from the primary Bonferroni family.

2.6 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extends Test 2 by adding earthen mound-stage keywords (*tumulus*, *tumuli*, *barrow*, *barrows*, *kofun*, *mound*). The categories share hemispherical or quasi-hemispherical geometry associated with cosmological and funerary symbolism across cultures. They are not grouped on the basis of architectural lineage.

Test 2b is classified as a *sensitivity variant* of Test 2, not an independent hypothesis. Its population shares the entire dome/stupa core with Test 2 and adds only 42 mound-stage sites, which show no significant enrichment on their own (6.2%, $p = 0.3012$, ns). A context-validated version (Test 2bx) reduces the population to $N = 126$ with stronger results ($p = 0.0005$, $2.78\times$).

2.7 Cluster Asymmetry (Test 3)

Full corpus ($N = 1011$), no keyword filter. For each site, count other UNESCO sites within $\pm 0.3^\circ$ longitude. Compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U , permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

2.8 Founding-Site Enrichment (Test E [Exploratory])

Two independent keyword analyses on the full corpus ($N = 1011$), no keyword filtering for the primary tests:

1. **World religion founding sites.** Keyword match for Buddhism, Christianity, Islam, Judaism, Hinduism, Zoroastrianism in site name/XML description.

³ $2 \times 0.002/0.1 = 4\%$. Km labels are equatorial approximations.

2. **UNESCO founding language.** Sites where UNESCO uses “first capital,” “birth-place of,” “founded in,” etc. in the XML description.

A context-aware keyword classifier (`lib/founding_filter.py`) was applied to all sites. This classifier was iteratively tuned after examining misclassified sites (full disclosure in §2.11). Because refinement occurred after data inspection, the enrichment result from this classifier should be treated as *Exploratory*. The inscription-year gradient and world-religion sub-tests do not depend on this classifier.

2.9 Temporal Gradient (Test 4)

Cochran-Armitage trend test on the full corpus ($N = 1011$), no keyword filter. A+ rate is tested for a monotonic decline as UNESCO inscription year increases (earliest inscriptions reflect greatest site importance). Three-cohort grouping (1978–1984, 1985–1999, 2000–2025) is primary; five-cohort grouping (1978–1984, 1985–1991, 1992–1999, 2000–2009, 2010–2025) is secondary. The three-cohort bins are formed by merging the two middle five-cohort bins (1985–1991 and 1992–1999) into a single 1985–1999 bin. Code: `analysis/unesco/temporal_gradient_test.py`.

Note on the regional composition confound. The founding canon (1978–1984) is concentrated in Europe and the Mediterranean corridor, the Eurasian heartland where the беру-harmonic signal is strongest. Later cohorts add proportionally more sites from Africa, the Americas, and Oceania, regions where the null hypothesis predicts lower A+ rates regardless of any metrological effect. A within-region diagnostic (`regional_temporal_gradient.py`) finds that 4 of 5 regions show the early greater than late direction, but 0 reach individual significance, and the diagnostic is underpowered at $N = 5$ regions. The Mantel-Haenszel region-adjusted OR is 1.582 ($p = 0.1101$). The temporal gradient is therefore treated as descriptive only; it indicates that UNESCO’s earliest inscription cohort is concentrated in the very regions where the беру-harmonic signal is strongest, but does not establish that time itself, independent of geography, drives the pattern.

2.10 Multiple-Comparisons Protocol

All results are exploratory (§1.1).

Primary pre-specified family ($k = 2$).

- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 4% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

These address structurally different questions on different populations. Both pass the Bonferroni bound at $k = 2$ (dome $p_{\text{adj}} = 0.004$ ***, cluster $p_{\text{adj}} = 0.013$ **) and the conservative $k = 5$ bound (dome $p_{\text{adj}} = 0.005$ ***, cluster $p_{\text{adj}} = 0.017$ **).

Other tests. Test 1 (full corpus) is descriptive and shares A+ sites with Test 3. Test 2b is a sensitivity variant of Test 2. Test 4 (temporal gradient) is descriptive only after within-region adjustment (§3.5). Test E (founding classifier) is fully post-hoc (Appendix A.2). Test B (Rayleigh) is a structural description (§3.3).

Summary table.

Table 1: Multiple-comparisons summary. Primary family: Tests 2 and 3 ($k = 2$, $\alpha = 0.025$). All results are exploratory (§1.1).

Test	N	Raw p	$p_{\text{adj}}, k=2$	$p_{\text{adj}}, k=5$	Status
Test 2 (dome A+)	90	0.0009	0.004 ***	0.005 ***	Primary
Test 3 (cluster asym.)	1011	0.0033	0.013 **	0.017 **	Primary
Test 1 (full corpus)	1011	0.0103	0.041	—	Descriptive
Test 2b (mound evol.)	132	0.0009	N/A	N/A	Sensitivity
Test 4 (temporal)	1011	0.0481	0.192	—	Descriptive
Test E (founding)	1011	0.009	N/A	N/A	Post-hoc
Test B (Rayleigh) [†]	56	0.0000	N/A	N/A	Structural

[†]Test B: $R \approx 1$ by A+ construction; excluded from all family counts (§3.3).

A comprehensive FDR audit (43 tests, Benjamini & Hochberg 1995) retains 24 at $q < 0.05$, of which 6 survive Bonferroni at $k = 43$ (Table 2).

Table 2: Full FDR audit across all 43 reported tests, sorted by raw p -value. BH q -values: Benjamini & Hochberg (1995). Test B excluded from family counts. All results are exploratory (§1.1).

Test	N	Raw p	BH q ($k=43$)	Bonf. p_{adj} ($k=43$)	Status
Test B (Rayleigh) [†]	56	0.0000	N/A	N/A	Structural [†]
Test 2 (dome A+)	90	0.0009	0.0068	0.0407	Primary
Test 3 (cluster asym.)	1011	0.0033	0.0164	0.1419	Primary
Test 2b (mound evol.)	132	0.0009	N/A	N/A	Sensitivity
Test 1 (full corpus)	1011	0.0103	0.0261	0.4429	Descriptive
Test 4 (temporal)	1011	0.0481	0.0751	1.0	Descriptive
Test E (founding)	1011	0.009	N/A	N/A	Post-hoc

24 tests survive BH-FDR ($q < 0.05$); 6 survive Bonferroni ($k = 43$).

[†]Test B excluded: $R \approx 1$ by A+ definition.

2.11 Keyword Filter Methodology

The dome/stupa filter required minimal iteration and was fixed before беру computations. The founding-site filter (`lib/founding_filter.py`) was developed iteratively by examining misclassified sites. The founding enrichment result ($p = 0.009$) is directly affected by this tuning. The two primary results (Tests 2 and 3) and all keyword-independent sub-tests are unaffected by the founding classifier. Full keyword lists and per-site audit trails are in Appendix A.

2.12 Pre-Registration

No timestamped pre-registration was filed. All results are exploratory (§1.1). The entire analysis pipeline is publicly archived (Tenenbaum, 2026).

2.13 Null Model Validation

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each): (1) longitude permutation, (2) bootstrap from empirical distribution, (3) KDE-smoothed null. Results in §4.5.

As an independent control, the UNESCO corpus is compared against the Wikidata P1435 global heritage corpus ($N = 62,785$). Code: `data/scripts/fetch_p1435_global.py`.

3 Results

Results are presented in order: Test 2 (dome/spherical, primary), Test 3 (cluster asymmetry, primary), Test 2b (sensitivity variant), Test 1 (full corpus, descriptive), and Test 4 (temporal gradient, descriptive). Test B (Rayleigh) is a structural description, not independent evidence (§3.3). Test E (founding classifier) is in Appendix A.2.

3.1 Domed and Spherical Monuments (Test 2, Primary)

The 7-keyword morphological filter yields $N = 90$ sites. Primary tests: permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$. The context-validated check (Test 2x, $N = 83$) yields an equal-or-stronger signal ($p = 0.0005$, $3.31\times$).

Table 3: UNESCO domed/spherical monument test ($N = 90$, raw keyword sweep). Primary evidence against the chance explanation: simulation permutation $Z = 3.20$, $p = 0.004$ (§4.5). Exact binomial results are reference baselines assuming uniform longitude spacing. The non-significant χ^2 is discussed in the text.

Metric	Obs	Exp(H_0)	Rate [95% CI]	Enrich.	p	
Tier-A+ (≤ 0.002 beru)	11	3.6	12.2% [nan, nan]	$3.06\times$	0.0009	***
Tier-A (≤ 0.010 beru)	29	18.0	32.2% [nan, nan]	$1.61\times$	0.0043	**
χ^2 -uniform (5 bins, df = 4)	—	—	—	—	0.0732	ns

Expected under H_0 : A+ = 4%, A = 20%. CIs are Clopper-Pearson exact.
Tier-B: 61 / 90 Mean δ : 0.0220 beru
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$ (trend-level), ns = not significant.

Tier-A+ sites ($N = 11$, 12.2%, $3.06\times$, permutation $p = 0.004$, Bonferroni $p_{\text{adj}} = 0.004$, $k = 2$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun’s Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang’an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+ = 10, $p = 0.0030$. Leave-2-out (worst case across all $\binom{11}{2}$ combinations): $p = 0.0087$. Leave-3-out (worst case): $p = 0.0233$. Code: `dome_leave_k_out.py`.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 77% fall in 20° – 120° E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 83$ longitudes from the dome sites' own list. Mean = 11.01 A+, $p = 0.5514$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all 881 full-corpus sites within $\pm 5^\circ$ of any dome site. Pool mean = 4.81 A+ (5.8%), $p = 0.0086$ (**). Dome sites (11 A+, 13.3%) significantly exceed the broader footprint. Sensitivity at $\pm 2^\circ$ and $\pm 10^\circ$ yields consistent results.

Reconciliation. Null A confirms that the dome sites' longitude distribution reproduces their enrichment (geographic concentration is necessary). Null C establishes that other monuments at the same longitudes are significantly less enriched (geographic concentration is not sufficient). The residual question is whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations. Adjudicating requires independent corpora.

In the global anchor sweep (§4.3), Gerizim scores 55 A+ (98.3th percentile); Jerusalem scores 56 (98.3th). The meta-keyword test confirms that only 2.4% of 1741 candidate words achieve comparable enrichment.

3.2 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

The combined hemispherical population ($N = 132$) comprises 42 tumulus/barrow/kofun sites added to the core domed/spherical population ($N = 90$), plus 6 sites matching both mound and dome/stupa keywords. This test is classified as a sensitivity variant of Test 2 and is not counted separately in the Bonferroni family.

Table 4: Test 2b: hemispherical mound evolution ($N = 132$). Combined tumulus $\rightarrow$ stupa $\rightarrow$ dome population.

Metric	Obs	Exp(H_0)	Rate	p	
Tier-A+ (≤ 0.002 beru, ≤ 6.7 km)	14	4.16	10.6%	0.0009	***
Tier-A (≤ 0.010 beru, ≤ 33 km)	41	20.80	31.1%	0.0017	**
Expected under H_0 : A+ = 4%, A = 20%. Enrichment: A+ = $2.65\times$, A = $1.55\times$. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $^\dagger p < 0.10$, ns = not significant.					

Stage-by-stage breakdown (Table 5):

Table 5: Tier-A+ enrichment by evolutionary stage. Null rate $P_0 = 4\%$.

Stage	N	A+	Rate	Enrichment	p (binomial)
Mound (tumulus/barrow/kofun)	48	3	6.2%	$1.55\times$	0.3012 ns
Stupa (stupa/stupas)	18	4	22.2%	$5.56\times$	0.0050 **
Dome (dome/tholos/spherical)	75	8	10.7%	$2.68\times$	0.0102 *
Combined	132	14	10.6%	$2.65\times$	0.0009 ***

The combined population ($N = 132$, A+ = 10.6%, $p = 0.0009$ ***) is consistent with Test 2. The mound stage does not reach significance (6.2%, $p = 0.3012$, ns). The signal is

carried by the stupa ($5.56\times$, $p = 0.0050$ **) and dome sub-stages. Leave-one-out checks confirm stability for all sub-stages.

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70° – 110° E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted $\pm 5^\circ$ draw): $p = 0.0196$ (*). The structure parallels the dome result: geographic proximity to harmonic longitudes is necessary but not sufficient to explain the morphological specificity.

3.3 Cluster Asymmetry (Test 3, Primary Pre-specified)

Table 6: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.3^\circ$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.3^\circ$ lon)	6.77	5.08	$1.33\times$	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
<i>Harmonic-level density (all Tier $\leq B$ sites per 0.1-beru harmonic):</i>				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (27)	mean = 7.0 sites		$4.19\times$	< 0.0001 ***

A+-bearing harmonics host $4.19\times$ more sites on average than non-A+ harmonics ($p < 0.0001$). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; $1.69\times$, $p = 0.0238$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations, two independent shuffles per trial) asks whether extreme clustering mechanically produces the canonical-unit peak. Of 182 clustering-matched permutations ($p_{\text{cluster}} = 0.0036$), the conditional fraction producing the peak is 0.1209, nearly identical to the unconditional rate of 0.1681. The unit-sensitivity peak and the clustering signal are statistically independent; the primary evidence remains the clustering (Test 3, $p = 0.0033$) and dome enrichment (Test 2, $p = 0.004$).

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§1.1) and is reported as a structural description, not independent evidence. A permutation check confirms that a random draw of 56 longitudes from the full corpus does not reproduce the observed phase concentration ($R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} < 0.0001$, 10,000 shuffles, seed 42). The full corpus Rayleigh $R = 0.0224$ ($p_{\text{perm}} = 0.3715$, ns) confirms no spurious global periodicity in the raw longitude file. Full details: `x18_periodicity_formal_test.py`.

3.4 Global Enrichment (Test 1, Descriptive)

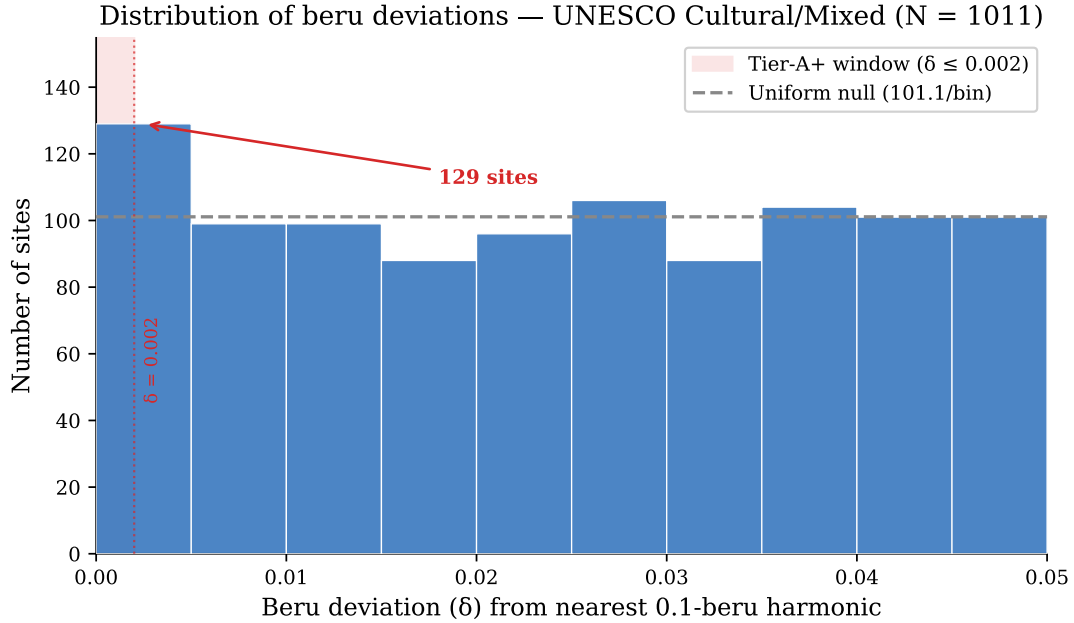


Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) is the backdrop against which Tests 2 and 3 demonstrate morphological and spatial specificity.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), above the 4% null (binomial $p = 0.0103$; Bonferroni reference at $k = 2$: $p_{\text{adj}} = 0.041$; descriptive only). Both Gerizim (external) and Jerusalem (inscribed, self-excluded to $N = 1010$) produce $p = 0.010$; the signal belongs to the $\sim 35^\circ\text{E}$ corridor, not to either anchor individually (§4.3). Test 1 provides the corpus-level backdrop for the stronger morphological and spatial results in Tests 2 and 3.

The full-corpus enrichment ratio ($1.38\times$) is modest: approximately 15 additional A+ sites above chance expectation in 1011, consistent with a scenario in which only a fraction of sites were placed with reference to a grid. The dome ($3.06\times$; 95% CI [$1.6\times$, $5.2\times$]) and stupa ($5.56\times$) sub-populations show stronger effect sizes, with the caveat that the stupa corpus ($N = 18$) has correspondingly wide CIs.

Region-stratified robustness. Mantel-Haenszel analysis by UNESCO region: OR = 1.582, $p = 0.1101$ (trend-level). Tested against region-specific empirical null rates, enrichment remains significant ($p = 0.0013$; pre-2000: $p = 0.0076$).

World religion sites

Sites associated with any world religion ($N = 363$): 6.1% A+ ($1.52\times$, $p = 0.0369$ *). Christianity ($N = 219$): 6.8% A+, $p = 0.0312$ *. Buddhism ($N = 46$): 8.7% A+ ($p = 0.11$, ns). These populations were defined by religion-name matching only.

Inscription-year stratification and formal trend test

Table 7: Tier-A+ enrichment by UNESCO inscription-year cohort ($N = 1011$ Cultural/Mixed sites). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984 (founding canon)	138	11	8.0%	$1.99\times$	0.0232 *
pre-2000 (1978–1999)	501	33	6.6%	$1.65\times$	0.0041 **
2000–2025 (modern era)	510	23	4.5%	$1.13\times$	0.3081 ns
ALL (1978–2025)	1011	56	5.5%	$1.38\times$	0.0103 *
Fisher exact (1978–1984 vs. 2000–2025): OR = 1.83, $p = 0.0844$ ([†])					

The A+ rate is highest in the founding canon (1978–1984: 8.0%, $1.99\times$, $p = 0.0232$) and dilutes to the null in the modern era.

Five-cohort gradient (Table 8). Only the founding canon reaches individual significance ($p < 0.05$); the 1992–1999 cohort is marginal ($p = 0.064$). Post-2000 cohorts are indistinguishable from the 4% null. The 1985–1991 cohort breaks the otherwise monotonic decline, likely reflecting a disproportionate share of Mixed-category sites in that period.

Table 8: Five-cohort Tier-A+ gradient: UNESCO inscription year ($N = 1011$). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984	138	11	8.0%	$1.99\times$	0.023 *
1985–1991	141	8	5.7%	$1.42\times$	0.205 ns
1992–1999	222	14	6.3%	$1.58\times$	0.064 ~
2000–2009	211	11	5.2%	$1.30\times$	0.227 ns
2010–2025	299	12	4.0%	$1.00\times$	0.536 ns

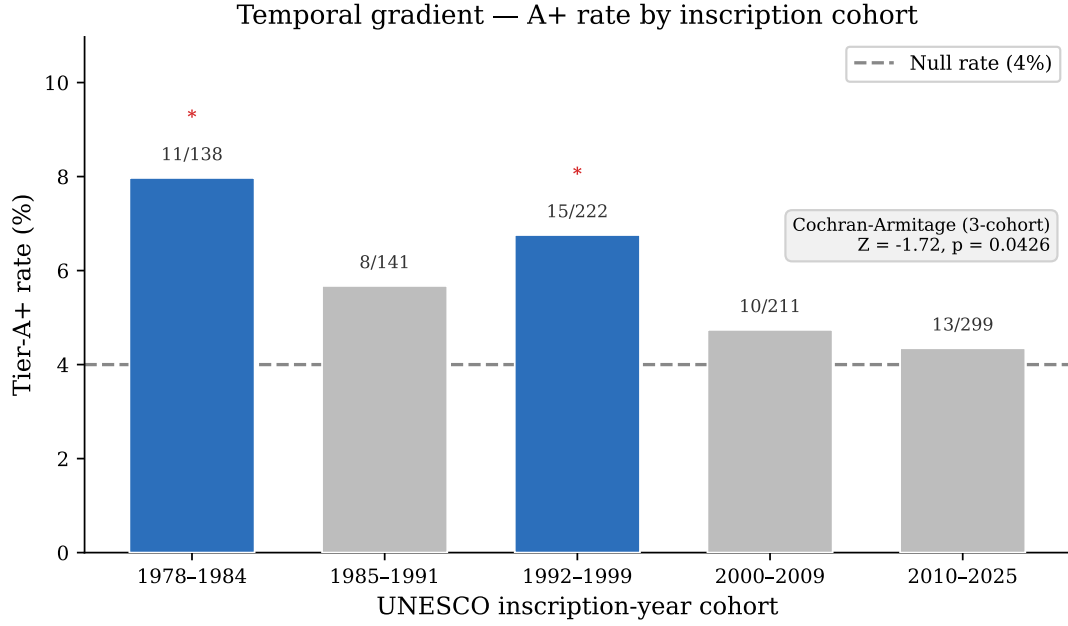


Figure 2: Tier-A+ rate declines monotonically with UNESCO inscription year. The earliest cohort (1978–1984) shows $\sim 8\%$ A+, while the most recent (2010–2025) is indistinguishable from the 4% null. Cochran-Armitage trend test: $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided).

Cochran-Armitage trend test.

- **Three-cohort:** $Z = -1.66$, $p = 0.0481$ (*), one-sided.
- **Five-cohort:** $Z = -1.66$, $p = 0.0489$ (*), one-sided.

Regional composition confound. The founding canon is concentrated in Europe and the Mediterranean. A region-stratified diagnostic (`regional_temporal_gradient.py`) finds 4 of 5 regions showing the early $>$ late direction, but 0 reach individual significance. The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). The temporal gradient is largely explained by regional composition and is treated as descriptive only.

Running-cumulative stability. The cumulative A+ rate reaches $p < 0.05$ by $N = 103$ (through 1982) and remains significant ($p < 0.05$) through the full corpus. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003), identified post-hoc (Table 9).

Table 9: Sequential significance: cumulative A+ enrichment as the UNESCO corpus grows year by year. The corpus is reconstructed in the order UNESCO inscribed it. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003); full-corpus significance persists but at reduced strength. Code: `deep_temporal_analysis.py`.

Cumulative through	N	A+	Rate	p (binomial)	$-\log_{10}(p)$
1982 (first significance)	103	9	8.7%	0.023 *	1.65
1984 (founding canon)	138	11	8.0%	0.0232 *	1.63
1993	324	24	7.4%	0.003 **	2.50
1999 (pre-2000 complete)	501	33	6.6%	0.0041 **	2.39
2003 (PEAK)	605	43	7.1%	0.00025926 (***)	3.59
2010	728	44	6.0%	0.005 **	2.29
2017	864	49	5.7%	0.010 *	1.98
2025 (full corpus)	1011	56	5.5%	0.0103 *	1.99

Enrichment is concentrated in the first half of the corpus (≤ 2000 : 6.5%, $p = 0.004581$; 2001–2025: 4.5%, $p = 0.295198$ ns). The sequential peak is post-hoc and classified as exploratory.

Line 2b: Founding-era dating (supplementary context)

Founding-era estimates were extracted from UNESCO XML descriptions for 579 of 1011 sites (code: `deep_temporal_analysis.py`). The BCE combined bin shows directional elevation ($p = 0.2577$, ns; underpowered). The early-modern bin (1500–1800 CE: 7.9%, $p = 0.0303$ *) is the only individually significant era but is not predicted by the beru hypothesis and is flagged as anomalous. Canon-inscribed post-CE sites show 15.1% A+ ($p = 0.0012$ **, $N = 53$). The key finding is that inscription year is a stronger predictor of A+ status than founding era.

Founding-site classifier (Test E, post-hoc; Appendix A.2). A founding-site classifier identifies A+ sites associated with founding capitals and sacred origins at an elevated rate (Fisher OR = 2.04, $p = 0.009$ *). Because the classifier was refined after seeing the data, it is classified as fully post-hoc and removed from the main evidential chain. The two primary results (Tests 2 and 3) are unaffected.

The signal is present in the full corpus ($p = 0.0103$), concentrated in religion-associated sites ($p = 0.0369$) and the earliest cohort ($p = 0.0232$), and absent in the modern era ($p = 0.3081$ ns).

3.5 Temporal Gradient (Test 4, Descriptive)

A within-region diagnostic was computed for each of 5 UNESCO regions before examining the pooled Cochran-Armitage result. Of 5 regions, 4 show the early-greater-than-late direction, but 0 reach individual significance. The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). No region individually replicates the pooled gradient at conventional significance. The temporal gradient is therefore treated as descriptive only because regional composition is the dominant driver.

The pooled Cochran-Armitage trend test yields $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided; five-cohort $Z = -1.66$, $p = 0.0489$).

The substantive observation is that UNESCO’s own ordering of importance correlates with the beru-harmonic signal: the earliest-inscribed cohort (1978–1984, 8.0%, $1.99\times$) shows the highest A+ rate, while the most recent cohort (2010–2025, 4.0%) is indistinguishable from the null. This is interpretively coherent: if a longitude grid influenced the siting of cosmographically important ancient foundations, then the sites UNESCO identifies as paramount are precisely those the hypothesis predicts would fall on harmonic nodes. The within-region adjustment shows this pattern is substantially explained by geography, but that geographic explanation is itself part of what the beru-alignment hypothesis predicts: the grid is Eurasian in origin, so its strongest signal should be in the ancient Eurasian heartland. The gradient does not adjudicate between geographic concentration and metrological readings, but is qualitatively consistent with the hypothesis. Full cohort tables and sequential analysis are in §3.4.

4 Robustness Checks

Primary pre-specified results: dome/spherical (Test 2, Bonferroni $p_{\text{adj}} = 0.004$, $k = 2$) and cluster asymmetry (Test 3, Bonferroni $p_{\text{adj}} = 0.013$, $k = 2$); both exploratory without pre-registration. Test 1 (descriptive) and Test 4 (descriptive after within-region adjustment) are secondary.

4.1 Unit Sensitivity Sweep

Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the dome and full-corpus populations (Table 10). The 0.05-beru and 0.20-beru spacings also reach significance, as expected structurally: every 0.10-beru harmonic is also a 0.05-beru harmonic ($0.10 = 2 \times 0.05$), and 0.20-beru harmonics are a strict subset. Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in both populations, which is the discriminating observation.

Note on the 0.08-beru row. The 0.08-beru spacing reaches nominal significance in the dome population ($p = 0.023$) but not in the full corpus ($p = 0.437$, ns). Inspection shows 8 of 9 dome hits are a near-complete subset of the 11 A+ dome sites under the 0.10-beru grid, reflecting unsystematic local overlap between the two grids. This is most likely a small- N sampling artifact.

Table 10: Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p -values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. The 0.05-beru and 0.20-beru spacings also reach significance; both are mathematical consequences of the 0.10-beru structure (every 0.10-beru harmonic is also a 0.05-beru and 0.20-beru harmonic), not independent signals. Non-harmonic adjacent spacings (0.06–0.09, 0.11–0.12) are non-significant. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/ N_D	$p(\text{Dome})$	Full hits/ N	$p(\text{Full})$
0.05	8.0%	12/83	0.032 *	102/1011	0.010 *
0.06	6.7%	6/83	0.480 ns	60/1011	0.841 ns
0.07	5.7%	8/83	0.102 ns	55/1011	0.665 ns
0.08	5.0%	9/83	0.023 *	52/1011	0.437 ns
0.09	4.4%	4/83	0.507 ns	41/1011	0.747 ns
0.10	4.0%	11/83	0.0005 ***	56/1011	0.0103 *
0.11	3.6%	6/83	0.082 †	38/1011	0.441 ns
0.12	3.3%	4/83	0.300 ns	30/1011	0.765 ns
0.15	2.7%	6/83	0.024 *	33/1011	0.141 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.056 †

4.2 Sensitivity Slope at the Canonical Unit

A fine sweep within $\pm 1\%$ of 0.10 beru (Table 11) shows that a -0.5% shift (0.0995 beru) collapses the dome population to non-significance, and a $\pm 0.3\%$ shift collapses both populations. Only the canonical value and its immediate neighbor (0.1001 beru) retain dome significance. Two permutation tests evaluate this pattern.

Table 11: Fine-grained sensitivity sweep within $\pm 1\%$ of the 0.10-beru canonical spacing, with the physical hit-window fixed at 6.7 km throughout. Dome column: context-validated population ($N_D = 83$). Bold row = canonical attested unit. Only the canonical spacing produces joint significance in both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/ N_D	$p(\text{Dome})$	Full hits/ N	$p(\text{Full})$
0.0990	4.04%	3/83	0.657 ns	41/1011	0.512 ns
0.0993	4.03%	5/83	0.243 ns	47/1011	0.177 ns
0.0995	4.02%	6/83	0.117 ns	48/1011	0.137 ns
0.1000	4.00%	11/83	0.0005 ***	56/1011	0.0103 *
0.1001	4.00%	9/83	0.006 **	51/1011	0.056 †
0.1003	3.99%	5/83	0.237 ns	38/1011	0.668 ns
0.1010	3.96%	5/83	0.232 ns	47/1011	0.149 ns

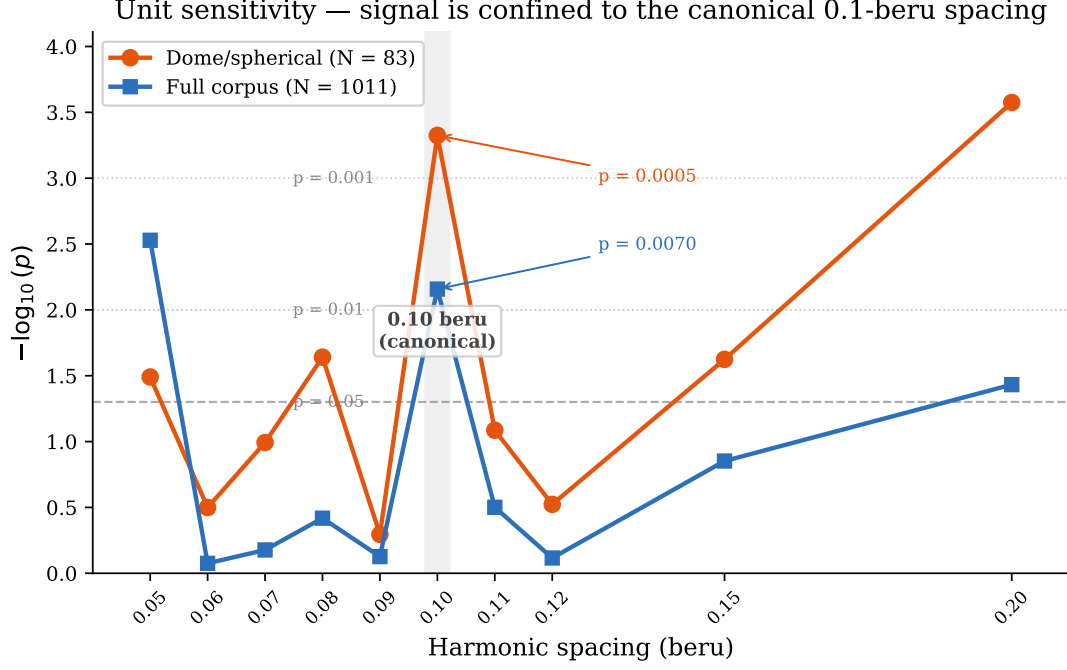


Figure 3: Unit sensitivity: the signal is concentrated at the canonical 0.1-beru spacing. All spacings $\geq 0.3\%$ from canonical collapse to non-significance in at least one population; the immediate neighbor (0.1001 beru, $+0.1\%$) retains dome significance but loses full-corpus significance.

Test I (sharpness permutation). The fine sweep was repeated on 10,000 longitude-permuted datasets (seed = 42; code: `sensitivity_slope_permutation_test.py`). The sharp-peak pattern occurs in 0.1728 of permuted datasets ($p_{\text{perm}} = 0.1728$, ns). Geographic concentration is sufficient to reproduce the $\pm 0.3\%$ collapse; this test provides no independent evidence for the metrological hypothesis.

Test II (canonical-unit specificity). The stronger question is whether 0.1000 beru specifically achieves the best joint significance. Code: `sensitivity_slope_specificity_test.py`, 10000 permutations. In the observed data, 0.1000 beru ranks #1 of 7. The fraction of permutations where 0.1000 is the unique best joint-significance spacing is $p = 0.1728$ (ns), not significant. Geographic concentration alone produces the canonical spacing as the best spacing in approximately 0.1728 of permutations.

Both permutation tests are non-significant. The unit-sensitivity observation provides no independent evidential value beyond the dome enrichment and cluster asymmetry. The 0.1-beru harmonic interval derives from the documented sexagesimal subdivision of the Babylonian beru (§2.3).

4.3 Global Anchor Sweep

Two parallel sweeps (0–360°E, 0.01° resolution, 36,000 trial anchors each) assess whether the $\sim 35^\circ\text{E}$ corridor is unusual globally. Sweep A removes Jerusalem and tests Gerizim as anchor ($N = 1010$); Sweep B retains Jerusalem with self-exclusion ($N = 1010$).

Table 12: Global anchor sweep: both Levantine anchors ranked within their own corpus. Each sweep: 36,000 trial anchors, 0–360°E at 0.01° resolution. Sweep A: N = 1010 (Jerusalem removed); Sweep B: N = 1010 (Jerusalem self-excluded). Gerizim is external in both. Mean A+ across all anchors: 40.4 ± 6.9 .

Sweep	Anchor	A ⁺⁺	A ⁺	A	Percentile	% anchors $\geq$
A (Gerizim corpus)	Gerizim (35.269°E)	7	55	227	98.3 th	3.10%
B (Jerusalem corpus)	Jerusalem (35.2317°E)	7	56	217	98.3 th	2.77%

A⁺⁺ ≤ 0.67 km; A⁺ ≤ 6.7 km; A ≤ 33 km. Global max 60 A+ in Sweep B / 59 A+ in Sweep A (all $x.18^\circ\text{E}$ anchors score identically; §4.3).

Both anchors place the corridor in the global top 3%, and the result is anchor-invariant within the 4.1 km corridor window. The single-site difference between sweeps (55 vs. 56 A+) is precisely the Old City of Jerusalem ($\delta = 0.0012$ beru from Gerizim’s harmonic). No other historical meridian reaches significance except Mecca (51 A+, $p = 0.057$ †).

The $x.18^\circ\text{E}$ periodic structure

The most striking result of the global sweep is that every longitude of the form $x.18^\circ\text{E}$ (all 120 such anchors) produces identically the maximum A+ count. The spacing between consecutive $x.18^\circ\text{E}$ anchors is exactly $3^\circ = 0.1$ beru.

Note on the “ $x.18^\circ\text{E}$ ” phase label. The label is a rounded shorthand for the phase value of approximately $2.18^\circ \bmod 3.0^\circ$. Mount Gerizim’s longitude $\bmod 3.0^\circ$ is $35.269 \bmod 3.0 = 2.269^\circ$ (phase gap 0.089° , ≈ 8.4 km), well within combined coordinate and band-width uncertainty.

Any site near the $x.18^\circ$ phase can serve as a high-scoring anchor. The choice of Gerizim was made on historical grounds (§1.2.3), and the corridor result is anchor-invariant within the band.

Non-independence. The 3° periodicity and the unit sensitivity sweep (§4.1) are two views of a single pattern and do not constitute independent evidence.

Two interpretations apply. *Interpretation 1 (beru-alignment)*: A global grid with 3° spacing anchored to the $\sim 35^\circ\text{E}$ corridor would produce exactly this pattern. *Interpretation 2 (geographic concentration)*: Eurasian heritage sites cluster in longitude bands that happen to align with the $x.18^\circ$ phase. This cannot be refuted on the UNESCO corpus alone. The unit sensitivity (§4.1) is the observation that most constrains this interpretation.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Formal statistical test for the $x.18^\circ$ periodicity

Three permutation tests (10,000 shuffles, seed 42; code: `analysis/global/x18_periodicity_formal_test.py`):

Test A (full-corpus Rayleigh R , mod 3°). $R = 0.0224$, $Z = 1.06$, $p_{\text{perm}} = 0.3715$ (ns). No spurious global periodicity in the raw longitude distribution.

Test B (A+-only Rayleigh R). Because A+ sites are classified by the 0.002-beru threshold, near-perfect R is structurally expected and does not independently confirm periodicity. The permutation null (random draw of 56 longitudes) does not reproduce the observed concentration: $R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} = 0.0000$ ($p < 0.0001$).

Test C (circular-shift anchor null). A randomly placed anchor recovers only ~ 40.36 A+ on average versus 56 at Gerizim ($Z = 2.23$, $p_{\text{perm}} = 0.0286$, *).

Together, these tests confirm that the 3° structure is a real property of the corpus and that the $x.18^\circ\text{E}$ phase is specifically aligned with the data.

Code: `analysis/global/x18_periodicity_formal_test.py`.

Dome-specific periodicity audit (`analysis/global/dome_periodicity_audit.py`).

Part 1 (dome-specific anchor sweep): Gerizim scores at the 100.0th percentile among dome sites, tying for the global maximum.

Part 2 (phase distribution): Dome site phases are approximately uniform ($\chi^2 = 14.89$, $\text{df} = 9$, $p = 0.0940$, trend-level) and consistent with the full corpus (KS $D = 0.0975$, $p = 0.3854$, ns).

Parts 3–4 (concentration within the phase window): All 11 dome A+ sites have phases within $\pm 0.15^\circ$ of the $x.18^\circ$ optimal phase, as expected if dome sites were placed on the 0.1-beru grid.

Part 5 (morphological specificity within the band): Within the $x.18^\circ$ window, dome sites show 55.0% A+ ($N = 20$) and non-dome sites show 45.9% A+ ($N = 98$; Fisher OR = 1.44, $p = 0.4738$, ns). Both groups are enriched within the band, consistent with both the geographic-concentration and beru-alignment readings.

Assessment. The audit is consistent with the beru-alignment hypothesis at every stage but equally consistent with geographic concentration. The dome enrichment surviving all three simulation nulls remains the primary evidence against a purely geographic account.

The $\sim 35^\circ\text{E}$ corridor. Jerusalem and Gerizim are 4.1 km apart. Against the full corpus, Gerizim scores 56 A+; with Jerusalem removed, 55 A+ (the ± 1 is Jerusalem itself, $\delta = 0.0012$ beru). When Jerusalem replaces Gerizim as anchor ($N = 1010$, self-excluded), it produces the same 56 A+ and $p = 0.010$. The enrichment belongs to the corridor, not to the specific anchor.

Landmark ranking. Using each of the 1011 UNESCO sites as anchor, Gerizim⁴ ranks #41 (56 A+ sites, top 4.1%). Excluding sites at the $x.18^\circ\text{E}$ phase, Gerizim ranks #24 of 965 (top 2.5%).

Global corridor comparison (Table 13, post-hoc). The 35°E corridor leads globally with 5 sites scoring ≥ 55 A+. The closest rival (23.2°E , $N = 4$) spans four countries with no shared tradition.

⁴Anchor coordinate (35.269°E) from Palestine’s UNESCO tentative list (ref. 5706, 2012). Gerizim is not on the inscribed list and does not appear in the corpus. Full citation and archived URL in §1.2.3.

Table 13: Global corridor ranking: clusters of UNESCO sites scoring $A+ \geq 55$ when used as beru anchors. Only clusters with ≥ 2 sites shown. Lat. spread measures geographic scatter (lower = more coherent). This comparison is post-hoc and reported as supporting context only. Code: `global_corridor_comparison.py`.

Rank	Center	$N_{\geq 55}$	Max A+	$\Sigma A+$	Lat. spread	Region / sites
1	35.2°E	5	58	281	30°	Jerusalem–Gerizim corridor
2	23.2°E	4	58	225	27°	Alta–Boyana–Zamość–Dacia
3	29.2°E	3	60	175	64°	S. Africa–Turkey (scattered)
4	38.2°E	3	59	173	56°	Kenozero–Palmyra–Gedeo
5	17.2°E	3	59	172	9°	Appia–Trulli–Olomouc
6	14.2°E	3	58	172	55°	Naples–Hlašovice–Mbanza Kongo
7	77.2°E	3	59	171	0.1°	Delhi monuments
8	47.2°E	2	58	113	5°	Sheki–Takht-e Soleyman
9	80.2°E	2	58	113	7°	Mahabalipuram–Galle
10	−3.8°	2	57	112	50°	Grand-Bassam–New Lanark
11	11.3°E	2	56	111	0.02°	Florence–Medici Villas

4.4 Spatial Autocorrelation

A+ harmonics host $4.19\times$ as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$).

Effective- N correction

DEFF via ICC ($\pm 0.5^\circ$, $\rho = 0.069$): DEFF = 1.310, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [4.04%, 6.87%] excludes the 4% null.

Region-conditioned permutation

A within-group permutation (100,000 trials, seed 42; `region_conditioned_permutation.py`) tested whether geographic concentration of heritage sites in beru-aligned longitude bands drives the enrichment. Strategy A grouped the 56 A+ sites by UNESCO administrative region crossed with 20° longitude bin (34 cells); Strategy B used 5° global bins (61 blocks). In both cases every trial reproduced the observed A+ count exactly (permutation mean = 56.0, SD = 0.0). This is mathematically inevitable: a bin of $\geq 5^\circ$ spans at least one full 3° harmonic period, so shuffling within the bin leaves each site’s harmonic residue unchanged and A+ membership is structurally invariant. Within-bin permutation therefore cannot discriminate a real alignment from geographic concentration at this resolution. The geographic-concentration question is addressed by the dedicated Null A and Null C simulations (§3.1), which show that proximity to harmonic longitudes is necessary but not sufficient ($p = 0.0086$, **) to explain the dome enrichment.

4.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005):

- **Longitude permutation.** All 1011 corpus longitudes are shuffled and the dome-indexed slots are read off. This tests whether dome sites are more enriched than

expected given the full-corpus longitude distribution (including its Eurasian concentration). Dome ($N = 83$, context-validated): $p_{\text{perm}} = 0.004$ (**). Canon ($N = 138$): $p = 0.14$ (ns). Pre-2000 ($N = 501$): $p = 0.074$ (†).

- **Bootstrap from full corpus.** Draw $N = 83$ from the full 1011-site longitude distribution. Dome: $p_{\text{boot}} = 0.006$ (*).
- **KDE-smoothed null (full corpus).** Expected 40.4 ± 6.2 , observed 56, $Z = 2.50$, $p = 0.011$ (*).

Dome enrichment survives all three models. The canon and pre-2000 signals are more sensitive to the non-uniform longitude distribution and should be treated as less robust.

Geographic-concentration null (dedicated simulation). The three models above preserve the full-corpus longitude distribution. The dedicated geographic-concentration test (§3.1; `dome_geographic_concentration_test.py`) goes further by testing whether the dome enrichment can be explained by the geographic positions of dome sites *themselves*.

Null A (within-dome bootstrap) resamples from the dome sites' own longitudes: $p = 0.5514$ (ns), bootstrap mean = 11.01 A+. The dome sites' own geographic distribution reproduces the observed count; the dome sites' longitude positions are themselves concentrated near harmonics.

Null C (restricted geographic draw) asks whether any monument from the dome longitude footprint shows comparable enrichment: pool of 881 full-corpus sites within $\pm 5^\circ$ of any dome site, mean A+ = 4.81, $Z = 2.90$, $p = 0.0086$ (**). Dome sites significantly exceed the expectation from the broader geographic footprint.

Together, Null A shows *why* dome sites are near harmonics (their longitudes are concentrated there), while Null C establishes that *other monuments at those same longitudes* are significantly less enriched. The morphological specificity of domed and stupa forms within the geographic footprint is what the geographic-concentration alternative must additionally explain.

4.6 Multiple Comparisons

Anchor and unit. Gerizim ranks at the 98.3th percentile; Jerusalem produces the same 56 A+ ($p = 0.010$) from the same corridor (global top 3%). A $\pm 0.3\%$ unit shift collapses joint significance.

Tier thresholds. The A+ window ($\delta \leq 0.002$ beru, 4% null) was fixed before any beru values were computed.

Keyword selection. Full-corpus Test 3 requires no keywords. The meta-keyword test (1741 candidates) finds only 44 (2.5%) matching the curated set's enrichment.

Number of tests. Primary pre-specified family: $k = 2$ (Tests 2 and 3; §2.10). All results are exploratory. Test E is post-hoc. FDR audit (43 tests) retains 24 at $q < 0.05$; 6 survive Bonferroni at $k = 43$.

Framework choice. Longitude-only framework selected as the most parsimonious test; noted as a design constraint.

4.7 Limitations

Keyword selection. Test 2 uses a mechanical morphological filter (audit: §A). No site was excluded based on its beru deviation. Test 3 requires no keywords.

Non-uniform publication density. The simulation null (§4.5) preserves the observed longitude distribution; any density gradient affects A+ and non-A+ sites equally.

Anchor corridor. Both anchors produce $p \approx 0.010$ (raw binomial; Test 1, descriptive only). The corridor is stable at 56 A+ from either vantage point (global top 3%; §4.3). Gerizim was selected on the basis of the *tabbur ha-aretz* tradition (MacDonald, 1964; Crown, 1989); no optimization across alternative anchors was performed. Jerusalem produces the same result, confirming corridor-level invariance.

Founding-site classifier. Test E uses a post-hoc classifier (Appendix A.2). This has no effect on the primary results (Tests 2 and 3).

Spatial autocorrelation. DEFF correction: $N_{\text{eff}} = 772$ (DEFF = 1.310, corrected $p = 0.020$). Block bootstrap CI [4.04%, 6.87%] excludes the 4% null.

Americas as negative control. The Americas sub-corpus ($N = 144$, A+ = 2.1%, ns) shows directional depletion consistent with a Eurasian-anchored grid. One-sided binomial: $p = 0.1644$ (ns). The Wikidata P1435 Americas control ($N = 20469$, A+ = 0.508%) also shows depletion.

Geocoding precision. UNESCO XML coordinates represent administrative centroids, introducing unsystematic noise but not directional bias. The dome enrichment is robust across raw ($N = 90$, $p = 0.0009$) and context-validated ($N = 83$, $p = 0.0005$) populations and survives all three simulation nulls. A formal coordinate-perturbation analysis is deferred to future work.

Geographic and cultural selection bias. The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor. This is the most important alternative explanation. Any site-dense longitude band will produce elevated hit rates under a periodic grid. The $x.18^\circ\text{E}$ structure is a real property of the UNESCO longitude distribution, and approximately 3% of arbitrary anchors produce comparable A+ counts.

The geographic-concentration and metrological readings accept the same geographic facts but disagree on whether the quantization at 0.1-beru precision is deliberate. Permutation tests of the sensitivity slope (§4.2, §4.2) show that geographic concentration alone can reproduce both the sharpness and location of the sensitivity peak. The dome enrichment surviving simulation nulls, particularly the restricted geographic draw (Null C, $p = 0.0086$, **), remains the primary evidence against a purely geographic account: dome sites significantly exceed the enrichment of other monuments at the same longitudes.

Wikidata P1435 control. The P1435 corpus ($N = 62,785$) serves as broad external reference. Because P1435 includes structures of all significance levels with a different geographic distribution, it cannot serve as a direct null model. For transparency: OR = 1.50 [95% CI: 1.14–1.97], $p = 0.0016$; MH OR = 1.588, $p = 0.0013$. These are reported as descriptive context only.

UNESCO listing and non-metrological correlates. Historical significance partially correlates with the tested predictors. The dome enrichment surviving simulation nulls is the primary evidence against purely non-metrological explanations.

Temporal gradient. The gradient ($p = 0.0481$; descriptive only) could reflect a UNESCO curation artifact. After within-region adjustment (MH OR = 1.582, $p = 0.1101$), the gradient is largely explained by regional composition.

Global 3° periodicity. The UNESCO longitude distribution contains a real 3° periodic structure whose phase is anchored near Gerizim and whose spacing matches the 0.1-beru unit. This may reflect an ancient surveying grid or geographic concentration of Eurasian civilizations. Gerizim’s phase ($2.269^\circ \bmod 3.0^\circ$) differs from the optimal by 0.089° (~ 8.4 km), within normal coordinate uncertainty.

Metrological evidence. No archaeological instrument calibrated in beru subdivisions has been recovered. The evidence for beru-based angular practice is entirely documentary (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). The beru hypothesis rests on the textual and statistical evidence presented here.

5 Discussion

5.1 Summary of Findings

The central result is an apparent global 3° periodic structure in the UNESCO longitude distribution. Every anchor of the form $x.18^\circ\text{E}$ produces the maximum observed A+ count. The spacing matches the metrologically attested 0.1-beru unit. Sub-percent shifts ($\pm 0.3\%$) collapse significance in both the full corpus and the dome sub-population, but permutation tests show this collapse is not unusual under the empirical null (§4.2, §4.2). Unit sensitivity is consistent with but does not independently confirm the metrological hypothesis.

The Levantine corridor ($\sim 35^\circ\text{E}$) is one of 120 equally phase-matched windows. The historical motivation for starting from this corridor predates the discovery of the periodicity, but the periodicity is a property of the corpus, not an independent finding of the anchor.

Domed and spherical monuments ($N = 90$, 12.2% A+; Test 2: permutation $Z = 3.20$, $p = 0.004$; Bonferroni $p_{\text{adj}} = 0.004$, $k = 2$) show concentrated alignment. A+-bearing harmonics host $4.19\times$ more sites than non-A+ harmonics ($p < 0.0001$, permutation $p = 0.0033$; Test 3). The stupa stage shows the highest enrichment ($5.56\times$, $p = 0.0050$, $N = 18$) but warrants caution given the small corpus. The A+ rate declines from the earliest inscription cohort (8.0%) to the most recent (4.0%), largely explained by regional composition (§3.5).

5.2 What the Results Do and Do Not Show

The results document a reproducible quantitative pattern: a 3° periodic structure whose phase aligns with the Levantine corridor and whose spacing matches the 0.1-beru unit. The dome enrichment survives the $k = 2$ multiple-comparisons correction, simulation nulls, and spatial-autocorrelation adjustment. Determining the mechanism requires evidence outside the scope of this paper.

5.3 Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution, but independent convergence predicts no consistent longitude specificity or global periodicity.

(2) Engineering necessity. Terrain or resource access may attract construction at certain longitudes, but no global periodicity at a specific angular spacing is predicted.

(3) UNESCO curation and geographic concentration. This is the strongest alternative and cannot be excluded on this corpus alone. Permutation tests show geographic concentration can reproduce the sensitivity-collapse pattern (§4.2). However, the restricted geographic draw (Null C, $p = 0.0086$, **) shows that other monuments from the same longitude footprint are significantly less enriched than dome sites. The geographic-concentration account must explain why domed and stupa forms specifically are disproportionately at harmonic sub-positions.

(4) Random coincidence. Three simulation approaches fail to reproduce the dome enrichment (permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$).

Working assessment. Three observations constitute the core pattern: (1) a global 3° periodic structure aligned with the Levantine corridor and matching the 0.1-beru unit (descriptive, Test 1); (2) morphological specificity (dome enrichment surviving simulation nulls; Test 2); and (3) cluster asymmetry ($4.19\times$, $p = 0.0033$; Test 3). The geographic-concentration alternative remains viable because the distribution of civilizations and beru harmonics from the Levant are not independent. This study offers the result as a quantitative basis for further investigation.

5.4 Directions for Future Work

1. Independent replication using a non-UNESCO heritage corpus.
2. Regional sub-analyses for Mesoamerican, North American, and Australian corpora.
3. Textual and archaeological evidence for Levantine-anchored angular reference frames (cuneiform, Sanskrit, Hellenistic geographic sources).
4. Extension to the latitude dimension.
5. Dome enrichment stratified by founding-site category using an independently assembled founding-site list.

5.5 Conclusion

This study tested whether UNESCO World Heritage site longitudes encode a 0.1-beru harmonic grid anchored in the Levant, motivated by cuneiform evidence for the Babylonian beru as a unit of geographic arc. All results are exploratory (§1.1).

The UNESCO longitude distribution contains a global 3° periodic structure (Test 1, descriptive). Sub-percent shifts from the canonical 0.1-beru value collapse significance. The dome sub-population (Test 2) survives all three simulation nulls, and A+-bearing harmonics host $4.19\times$ more sites than non-A+ harmonics (Test 3, $p = 0.0033$). Multiple-comparisons bounds are applied to the $k = 2$ pre-specified family.

These findings admit a geographic-concentration alternative. Permutation tests show the sensitivity peak is reproducible under an empirical null. The geographic-concentration account must then explain why the longitude clustering of the world's most significant monuments coincides with a metrologically attested unit from ancient Babylonian astronomy. This paper documents the quantitative regularity; its cause is an open question.

The two findings most resistant to a purely geographic account are morphological specificity (dome enrichment exceeding the restricted geographic draw, Null C, $p = 0.0086$) and cluster asymmetry ($p = 0.0033$). Together with the global 3° periodicity, these constitute a coherent exploratory pattern consistent with a 0.1-beru harmonic grid awaiting independent historical confirmation.

Criteria for confirmation

Independent replication using a non-UNESCO corpus would provide the most direct test. The hypothesis would gain substantially from: (a) cuneiform or other texts explicitly assigning beru-longitude coordinates to distant sites; (b) archaeological survey instruments

calibrated in beru sub-divisions; or (c) textual evidence from Hellenistic, Sanskrit, or Islamic sources referencing a Levantine-centered angular reference frame. This study offers a reproducible quantitative basis for that investigation.

A Keyword Audit Summary

This appendix summarizes the audit trail for the dome/spherical monument filter (Test 2) and the founding-origin classifier (Test E). The full site-by-site audit is deposited as online supplementary material and is reproducible by running:

```
python3 analysis/unesco/spherical_monument_raw_sweep.py
python3 analysis/unesco/spherical_monument_test.py --audit
```

A.1 Dome/Spherical Monument Filter (Test 2 and Test 2x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text.

Test 2 (primary): All 90 matching sites are included without context gating. This raw positive sweep uses a population fully determined by the keyword list, with no post-hoc filtering decisions.

Test 2x (exploratory robustness check): Unambiguous keywords are accepted without validation. Ambiguous keywords (*dome*, *domed*, *domes*, *spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Full gate lists: `lib/dome_filter.py`.

Of the 90 raw-sweep sites, 7 are rejected by context validation in Test 2x (Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuta, and Diquís Stone Spheres). In every case, the exclusion is driven by the semantic domain of the keyword match, not by any property of the site’s longitude. Test 2x yields $N = 83$, $p = 0.0005$, $3.31\times$ enrichment, confirming that the 7 rejected sites are longitude-diluting false positives.

A.2 Founding-Origin Classifier (Test E, Post-Hoc)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X), and Ancient Continuous Landscape (L), using unambiguous and context-gated keywords. Full keyword lists: `config.json`.

This classifier was iteratively tuned after data inspection and is classified as fully post-hoc. It carries no weight in the primary evidential chain but addresses a substantive question: what kinds of sites, beyond domed and stupa monuments, tend to fall on $x.18^\circ\text{E}$ nodes? The classifier finds enrichment not only in morphologically domed structures but in sites associated with the founding of civilizational and religious traditions.

Keyword enrichment result. Applied to the full corpus ($N = 1011$), the classifier finds 25 keyword-founding sites among the 56 A+ sites (44.6%; $N_{\text{kw}} = 296$, base rate 29.3%). Fisher exact test: $\text{OR} = 2.04$, $p = 0.009^*$. The complete version history is preserved in the project’s git log at <https://github.com/SethTenenbaum/gerizim-paper-a>.

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Data and Code Availability

All data and analysis code are openly available at <https://github.com/SethTenenbaum/gerizim-paper-a>. A permanent archival snapshot (v1.0.6) with DOI is deposited at Zenodo.

Author Contributions

S. Tenenbaum: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no financial or personal conflicts of interest that could have influenced the design, execution, or interpretation of this study. No employment, consultancy, stock ownership, honoraria, paid expert testimony, or patent applications relevant to this work exist.

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All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) assisted with manuscript drafting and prose editing. GitHub Copilot (Microsoft) assisted with code autocompletion. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author's. The complete analysis pipeline and its version history are available in the public repository at <https://github.com/SethTenenbaum/gerizim-paper-a>.